ESP32-S3-DevKitC-1 native USB path

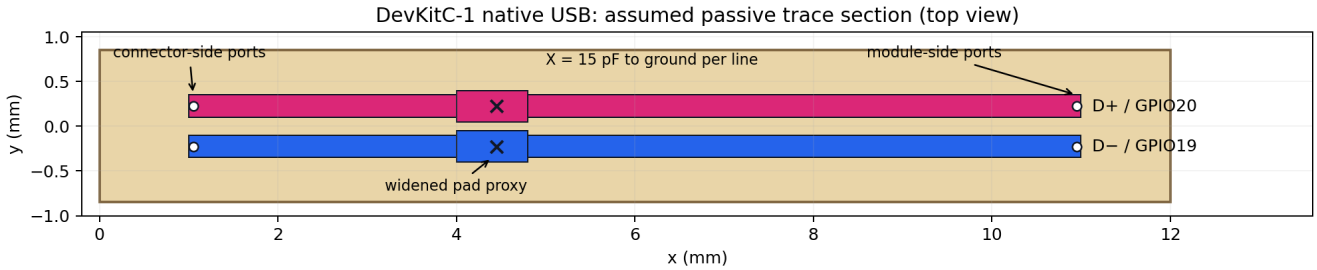
An openEMS model of one passive PCB interconnect section. It is not a simulation of the ESP32-S3 chip or the complete development board.

What comes from the board

Espressif's v1.1 schematic connects connector J4 D- to GPIO19 and D+ to GPIO20. Protection diodes D8 and D9 connect those signals to ground. The indicated part is LESD5D5.0CT1G; its datasheet lists 15 pF typical junction capacitance at 0 V and 1 MHz. The loaded run replaces each diode with a fixed 15 pF shunt capacitor. This is a linear small-signal approximation, not ESD clamping behavior.

What the geometry assumes

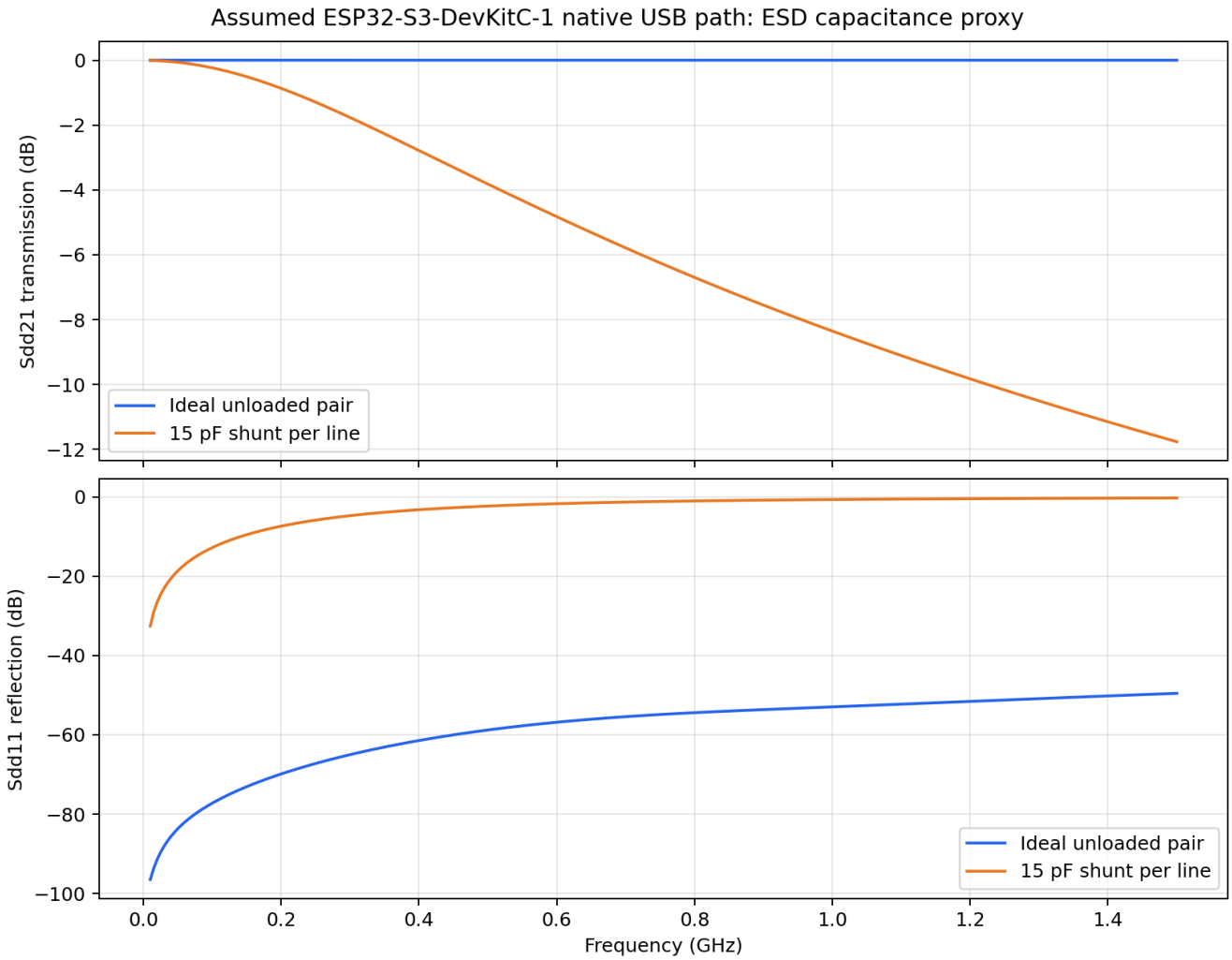
Quantity	Model value	Status
Straight pair length	10 mm	Assumed
Each trace width / gap	0.25 mm / 0.20 mm	Assumed
Dielectric to ground	0.20 mm, er = 4.2	Assumed
Copper / dielectric loss	Ideal copper, lossless FR-4	Simplification
Port reference	50 ohm per line (100 ohm diff)	Chosen for simulation
ESD capacitance	0 or 15 pF per line	15 pF from part datasheet



The published PCB layout was not imported. Pad shapes and all route dimensions above are illustrative. Use the XML in AppCSXCAD to rotate and inspect the 3D model.

Two electrical runs

Both runs use the same ideal trace geometry and opposite-polarity drives on D+ and D-. One run has no diode load; the other has a 15 pF shunt on each signal.



Reading the result

At 10 MHz, Sdd21 is -0.001 dB unloaded and -0.010 dB with the capacitance proxy. Near 1 GHz, it is -0.001 dB unloaded and -8.365 dB loaded. The fixed shunt capacitance increasingly loads fast spectral components. These frequency-domain curves alone do not establish a USB eye diagram or USB compliance.

Solver status

Unloaded: Field decay target reached after 32,186 steps (-51.73dB).

15 pF proxy: Field decay target reached after 33,022 steps (-50.92dB).

A reached field-decay target does not prove mesh convergence. These first-mesh numbers are exploratory. Exact board prediction requires the routed copper, stackup, diode package/parasitics, connector, chip input model, and convergence checks.

Run and inspect

From simulations/: `./scripts/run_devkitc1_usb.sh`; `./scripts/run_devkitc1_usb.sh --case devkitc1_usb_esd15pf --esd-cap-pf 15`; `./scripts/view_devkitc1_usb.sh`.

Sources

Espressif DevKitC-1 v1.1 guide | Espressif schematic | Leiditech component datasheet